

AKSHAY ASHOK

M.Sc. Data Science & Artificial Intelligence | Saarland University

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PROFILE

Master's student in Data Science & AI (5th semester) who ships measured, reproducible ML: a first-author Springer publication in computer vision, a reproducibility study of a multimodal LLM that tests what its authors never did, an Arabic-to-German OCR/translation benchmark and a deployed NLP scoring service. Strongest in evaluation design and failure analysis: finding where a reported result does less work than it appears to. Open to a Master's thesis topic from late 2027 and research collaborations.

EXPERIENCE

Working Student, Data Science & Process Intelligence · ORBIS Value+ · Saarbrücken Oct 2026 – Present

- Data science in the Process Intelligence team, part-time alongside the M.Sc.

Student Consultant (AI / Data Science) · Lingomatch GmbH · Saarbrücken Apr – Jun 2026

- Built and benchmarked the ML backend of a document-triage feasibility study for a Gov/Justice-Tech language-services platform (UdS Innovation Center, Triathlon case-study programme): turning photos of Arabic documents into usable German gist translations within an 8 GB laptop-GPU budget.
- Evaluated Surya OCR on KITAB-Bench (per-sample CER/WER, confidence calibration); tested PaddleOCR as an independent second engine for disagreement-based flagging and documented why it fails on Arabic.
- Benchmarked Opus-MT vs. NLLB vs. TranslateGemma for translation; caught the licence issue (CC-BY-NC) ruling the best baseline out of production and recommended an Apache-2.0 model with comparable BLEU at 1/9 the memory.
- Designed confidence-based failure routing that flagged 4/4 catastrophic OCR failures, defining the pipeline's human-escalation path; presented the live demo in the final case-study presentation to the company's CEO.

github.com/Axwolf13/arabic-doc-triage-research · write-up: axwolf13.github.io/writing/arabic-ocr-benchmarks

Research Intern (Computer Vision) · National Institute of Technology, Puducherry Jan – Apr 2024

- Built a real-time yoga pose estimation pipeline with a hybrid MoveNet Thunder (training) + MoveNet Lightning (inference) architecture: 11 asanas at 98.8% test accuracy (98.4% verified on the shipped quantized model) at 30+ FPS on consumer hardware; curated the 3,435-image dataset end to end.
- First-authored a peer-reviewed publication in *Sādhana* (Indian Academy of Sciences, Springer, 2025); released the code, the trained model and a live in-browser demo (TensorFlow.js) with a one-command verification script.

doi.org/10.1007/s12046-025-02788-w · live demo: axwolf13.github.io/demo

Web Development Intern · Zoho Corporation, Chennai, India Aug – Sep 2023

- Built a full-stack library management web app with a content-based book recommender (HTML/CSS, JavaScript, SQL).

SELECTED PROJECTS

MolCA reproduction: which input does a molecule LLM believe? · graduate seminar, Prof. Klakow

- Reproduced MolCA (EMNLP 2023) on two machines: 62.32 and 62.77 BLEU-2 against the paper's 62.0, 93.8% of captions character-identical, all 32 published retrieval metrics within 0.5 points; 14 patches to the code, several for unreported bugs.
- Designed the modality-conflict test the paper never ran: given one molecule's SMILES and another's graph, the model follows the graph about 90% of the time; a wrong graph costs 2.5× what wrong text does (paired bootstrap, every interval excludes zero); controlled for 49% train-test contamination on a second dataset.

github.com/Axwolf13/molca-reproduction · write-up: axwolf13.github.io/writing/molca

DODI: deception scoring for Terms-of-Service documents, shipped as a live web service

- Deterministic index of how hard a ToS hides that "Buy now" means "revocable licence"; temporal study of 10 platforms across 2015–2024, compared against ToS;DR expert grades.
- Cross-checked with three LLM judges from two vendors (rubric-based structured scoring, self-consistency repeats): they barely agree with the index ($\rho = 0.12$, $n = 51$), exposing a false-positive mode. Released v1.1 with whole-word matching and a reviewed-grades validation set.
- Deployed as a Dockerized FastAPI service with tests and CI (dodi-web.onrender.com) plus an MCP server over Streamable HTTP, tested as a tool in a Microsoft Copilot Studio agent.

EDUCATION

M.Sc. Data Science & Artificial Intelligence · Saarland University Oct 2024 – Present

Coursework: Statistics with R, Human-Computer Interaction, Data and Society. Seminar: ML for Language Processing (MolCA).

B.Tech. Artificial Intelligence & Data Science · Bannari Amman Institute of Technology Oct 2020 – Apr 2024

CGPA: 8.65/10. Thesis: LipNet-based visual speech recognition as a personal communication aid for hearing-impaired users.

TECHNICAL SKILLS & LANGUAGES

Languages & tools: Python, R, SQL, JavaScript; PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, spaCy, OpenCV, pandas, NumPy; FastAPI, Docker, CI/CD (GitHub Actions), Git, HTCondor, LaTeX.

LLM work: OpenAI, Anthropic & Gemini APIs, prompt design, structured outputs, LLM-as-judge evaluation, MCP server development, embedding-based retrieval, agentic workflows (Claude Code, Cursor).

Methods: model benchmarking and evaluation design (BLEU, CER/WER, calibration, paired bootstrap CIs), reproducibility and ablation studies, classical ML & deep learning, mixed-effects models, event-study design, NLP and OCR pipelines.

Spoken languages: English (C1), German (A2, targeting B1), Tamil (native), Hindi (basic).